

Yidi Zhang

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Education

Zhejiang University, Chu Kochen Honors College Sept. 2022 – June 2026
Bachelor of Engineering, Computer Science and Technology (Expected)

Advisor: Prof. Sida Peng

- GPA: 4.12/4.30 (91.27/100)
- **Selected Coursework:** Mathematical Analysis I(H) (100/100), Linear Algebra II(H) (99/100), Introduction to Natural Language Processing (97/100), Advanced Data Structures and Algorithm Analysis (94/100)

University of California, Berkeley Jan. 2025 – May 2025
Exchange Student

- **Selected Coursework:** COMPSCI 280: Computer Vision (Graduate Level) (A)

Publications

RobotArena ∞ : Scalable Robot Benchmarking via Real-to-Sim Translation

International Conference on Learning Representations (ICLR), 2026

Yash Jangir, **Yidi Zhang***, Kashu Yamazaki*, Chenyu Zhang*, Kuan-Hsun Tu, Tsung-Wei Ke, Lei Ke, Yonatan Bisk, Katerina Fragkiadaki

Project | arXiv

Experience

Tencent AI Lab Jan. 2026 – Present
Research Intern

- Researching and developing **Generalist GUI Agents** capable of navigating complex software ecosystems by treating digital interfaces as semi-structured environments for perception and planning.

Carnegie Mellon University Jan. 2025 – Jan. 2026

Research Intern (On-site: June – Sept. 2025)

Advisor: Prof. Katerina Fragkiadaki

- **RobotArena ∞ : Scalable Robot Benchmarking via Real-to-Sim Translation**—Developed a scalable benchmark for real-world-trained robot policies, converting video demonstrations into simulated environments with automated vision-language scoring and human preference feedback, enabling reproducible, robust, and large-scale evaluation of robot generalist performance.
- Led development of the object pose recovery pipeline and human preference evaluation framework, enabling accurate real-to-sim reconstruction and unbiased policy comparison across thousands of demonstrations.

Zhejiang University Aug. 2023 – Jan. 2025
Research Intern, Advisors: Prof. Sida Peng & Prof. Xiaowei Zhou

- Developed a solid foundation in 3D computer vision through systematic study of NeRF, 3D Gaussian Splatting, and differentiable rendering techniques.
- **4D Gaussian Splatting Research**—Improved the optimization framework with depth-aware supervision for better spatio-temporal consistency and geometric fidelity, and enabled efficient streamable rendering with significantly reduced memory usage.

Projects

DREAMoR: Diffusion-based Reconstruction and Motion Prior Project Page

- Final project for UC Berkeley COMPSCI 280: Computer Vision (Graduate Level).
- Proposed **DREAMoR**, a diffusion-based motion prior framework for reconstructing physically plausible human motion from monocular sequences degraded by occlusion or noise.
- Developed a latent-space diffusion prior with multi-step DDIM denoising and score distillation

optimization, integrating temporal consistency and physics-based constraints to enhance realism and robustness.

SpatialPi: 3D-Enhanced Robot Learning with Point Cloud Representations

Project Page

- Developed a comprehensive toolkit that constructs 3D point cloud representations from robot demonstrations and fine-tunes vision-language-action models to improve spatial reasoning in robotic manipulation.

3D Scene Inpainting with ControlNet

GitHub

- Developed a PyTorch implementation of ControlNet-based inpainting models for completing missing regions in 3D scene images, supporting both Stable Diffusion 1.5 and SDXL architectures.
- Trained models from scratch and fine-tuned pretrained ControlNet on custom 3D scene datasets, achieving realistic scene completions through mask-conditioned diffusion.

Chain-of-Thought Reasoning with Large Language Models

GitHub

- Developed and implemented innovative techniques including Multi-vote, LoRA fine-tuning, and Multi-path step-aware inference to enhance reasoning and problem-solving capabilities of large language models.
- Designed a comprehensive experimental framework to evaluate and compare different chain-of-thought prompting strategies.

Volunteer Experience

Volunteer Teacher

Summer 2023

Tacheng, Xinjiang, China

- Volunteered to teach in one of China's most remote regions, introducing students to science and the wider world through hands-on experiments and storytelling.

Skills

Programming: Python, C/C++ , RISC-V Assembly, Bash, Verilog, HTML

Frameworks: PyTorch, Open3D, OpenCV

Tools: Git, COLMAP, MeshLab, $\text{\LaTeX}$

Languages: Chinese (Native), English (Fluent)

Honors & Awards

First-Prize Scholarship of Zhejiang University

Fall 2024

- Awarded by Zhejiang University, Top 2%

Scholarship for Exceptional Students in Fundamental Subjects

Fall 2023

- Awarded by Zhejiang University, Top 10% in department